

Input:

- V . Set of nodes
- E_s . Set of scheduled edges (v, u, dep, arr)
- E_t . Set of transfer edges (v, w, Δ)
- $country(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets
- $bucket(t)$. Time bucket mapping function

Output:

- P . Sequence of labels representing a route within 24 hours

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1: Initialize  $Labels(v) \leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize queue  $Q \leftarrow \emptyset$ 
3: Initialize  $best\_label \leftarrow \text{null}$ 
4: // Initialize labels at departures
5: for all  $v \in V$  do
6:   for all scheduled departure from  $v$  at time  $dep$  with arrival  $arr$  do
7:      $L \leftarrow (v, dep, \{country(v)\}, \text{pred} = \text{null}, \text{start\_time} = dep)$ 
8:      $Labels(v) \leftarrow Labels(v) \cup \{L\}$ 
9:      $Q \leftarrow Q \cup \{L\}$ 
10: // Main loop
11: while  $Q \neq \emptyset$  do
12:   extract  $L = (v, t, C, \text{pred}, \text{start\_time})$  from  $Q$ 
13:   // Enforce 24-hour journey limit
14:   if  $t - \text{start\_time} > 24 : 00$  then
15:     continue
16:   // Upper bound pruning
17:    $b \leftarrow bucket(t)$ 
18:    $UB \leftarrow |R(v, b) \setminus C|$ 
19:   if  $best\_label \neq \text{null} \wedge |C| + UB \leq |best\_label.C|$  then
20:     continue
21:   // Update best label
22:   if  $best\_label = \text{null} \vee |C| > |best\_label.C|$  then
23:      $best\_label \leftarrow L$ 
24:   // Extend via scheduled edges
25:   for all  $(v, u, dep, arr) \in E_s$  do
26:     if  $dep \geq t \wedge dep - \text{start\_time} \leq 24 : 00$  then
27:        $t_{\text{new}} \leftarrow arr$ 
28:       if  $t_{\text{new}} - \text{start\_time} > 24 : 00$  then
29:         continue
30:        $C' \leftarrow C \cup \{country(u)\}$ 
31:        $L' \leftarrow (u, t_{\text{new}}, C', \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
32:       if  $\nexists L'' \in Labels(u)$  such that
33:          $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C' \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C'))$  then
34:           remove all  $L'' \in Labels(u)$  such that
35:              $(t_{\text{new}} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C' \neq L''.C))$ 
36:            $Labels(u) \leftarrow Labels(u) \cup \{L'\}$ 

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35:   |   |   |    $Q \leftarrow Q \cup \{L'\}$ 
36:   |   |   |   // Extend via transfers
37:   |   |   |   for all  $(v, w, \Delta) \in E_t$  do
38:   |   |   |   |    $t_{\text{new}} \leftarrow t + \Delta$ 
39:   |   |   |   |   if  $t_{\text{new}} - \text{start\_time} > 24 : 00$  then
40:   |   |   |   |   |   continue
41:   |   |   |   |    $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
42:   |   |   |   |   if  $\nexists L'' \in \text{Labels}(w)$  such that
43:   |   |   |   |   |    $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$  then
44:   |   |   |   |   |   |   remove all  $L'' \in \text{Labels}(w)$  such that
45:   |   |   |   |   |   |    $(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$ 
46:   |   |   |   |   |   |    $\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$ 
47:   |   |   |   |    $Q \leftarrow Q \cup \{L'\}$ 
48:   |   |   |   // Reconstruct path
49:   |   |   |    $P \leftarrow$  empty sequence
50:   |   |   |    $L \leftarrow \text{best\_label}$ 
51:   |   |   |   while  $L \neq \text{null}$  do
52:   |   |   |   |   prepend  $L$  to  $P$ 
53:   |   |   |   |    $L \leftarrow L.\text{pred}$ 
54:   |   |   |   return  $P$ 

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In Algorithm 2 we make a change that targets one of the main issues in Algorithm 1: the explosion of labels, which are rarely pruned due to the very relaxed conditions of dominance.

As seen in the notes, we use the Upper Bound of reachable countries from each node at a specific time (bucket) to prune hopeless results that would otherwise stay alive.